Building Better Context with Visual Intelligence & Semantic Retrieval

From messy data to relevant answers with Jina models on Elastic Inference Service

elastic



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ML Engineer II

Inference Team

User Story

- Meet Jane
- Archivist at Big Museum in London
- Role
  - **Transcribe** historical documents
  - **Store** them in a database
  - **Search** database for most relevant ones
- Messy data
- Long hours!

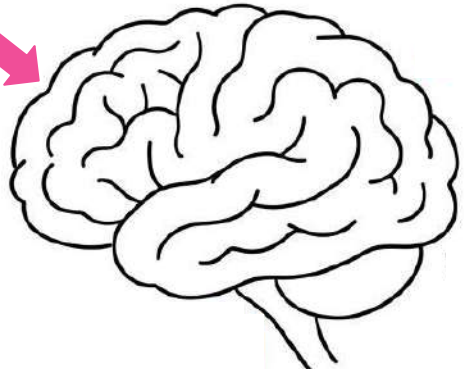
Just ask an LLM to do it! NO

- Cost
- Energy
- Inefficient



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LLM



What can Jane do instead?

Use small models to give LLMs better context

Embeddings v5

Reranker v3.5

OCR v1

Pipeline

Transcribe docs

Extract

- Jina OCR

Write to database

Ingest

- Jina Text Embeddings v5

Search database

Search

- Jina Text Embeddings v5
- Jina Reranker v3.5

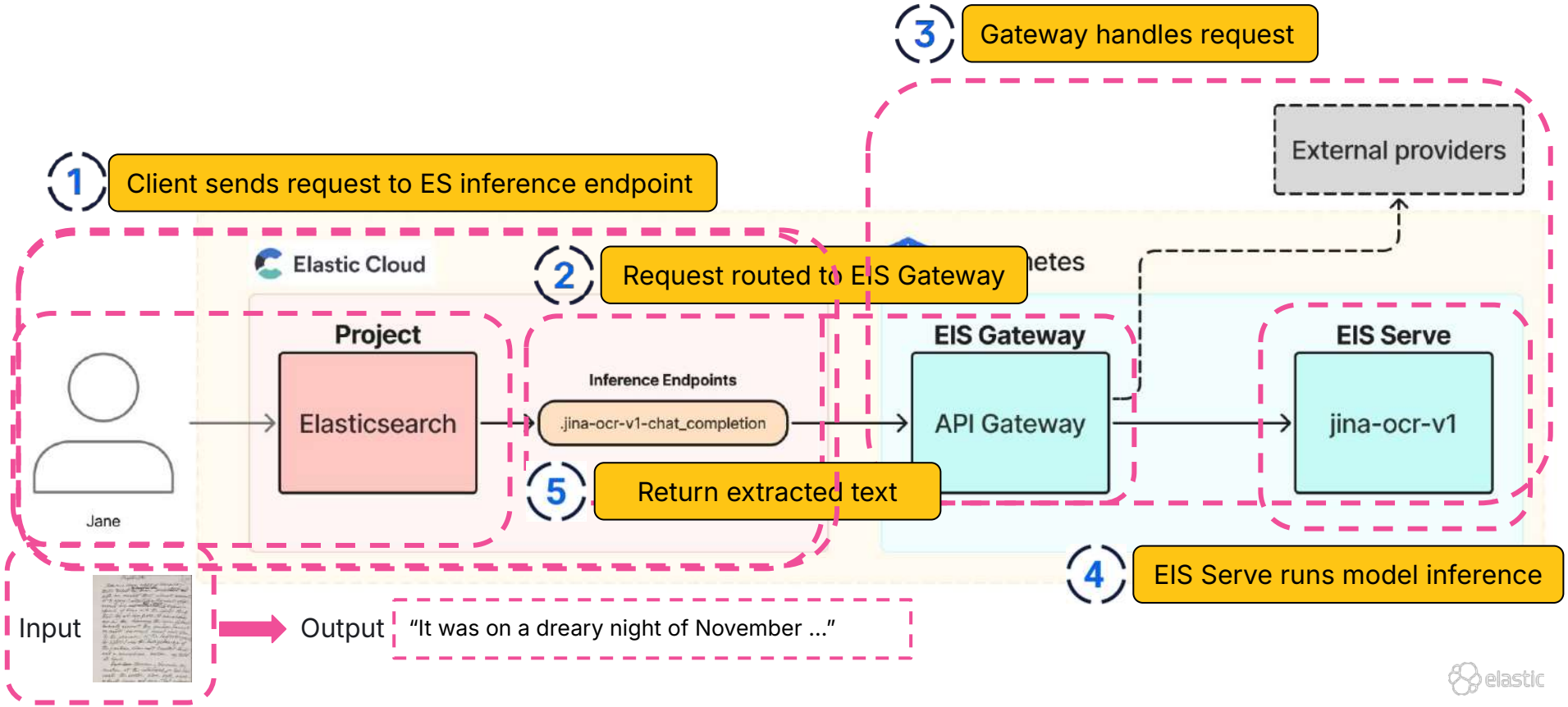
Elasticsearch

+

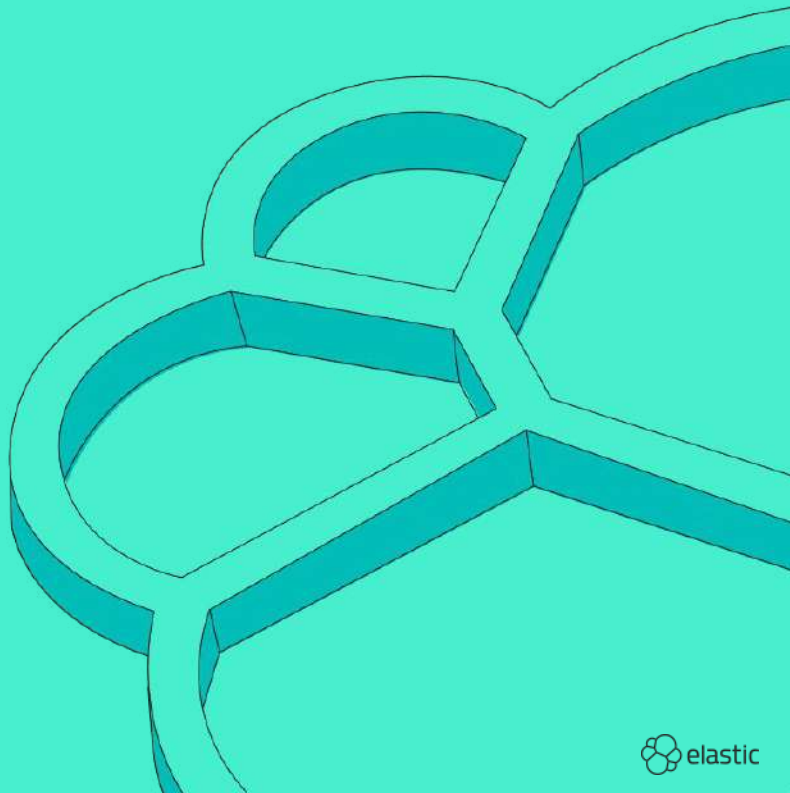
Elastic Inference Service (EIS)

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Diagram - Extract



Demo: Extract

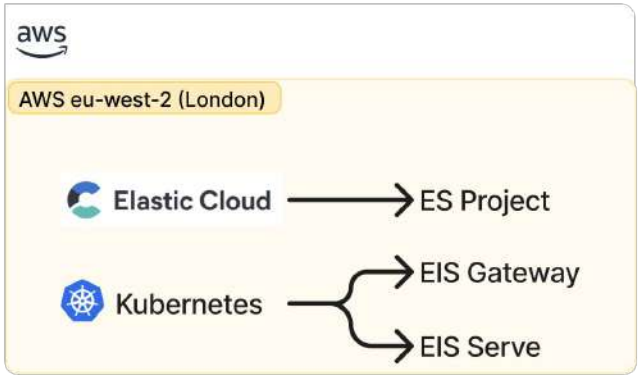


Key points

Infrastructure

Same CSP and Region

- Data residency
- Closer = Low Latency

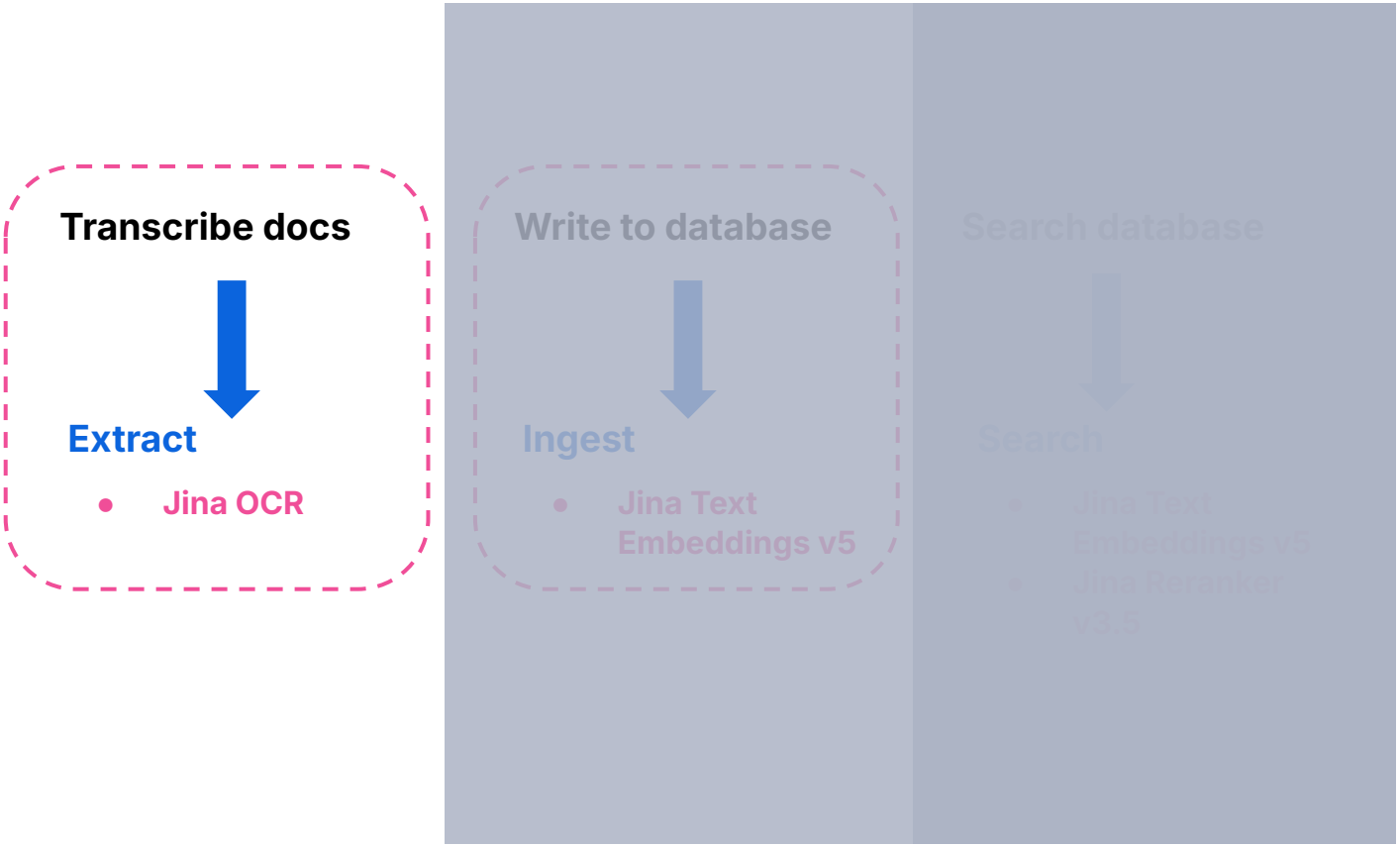


Self-managed

- Write once use everywhere
- Maximum control
  - Scaling
  - Hardware
  - Drivers
  - Observability
- In-house infra tool

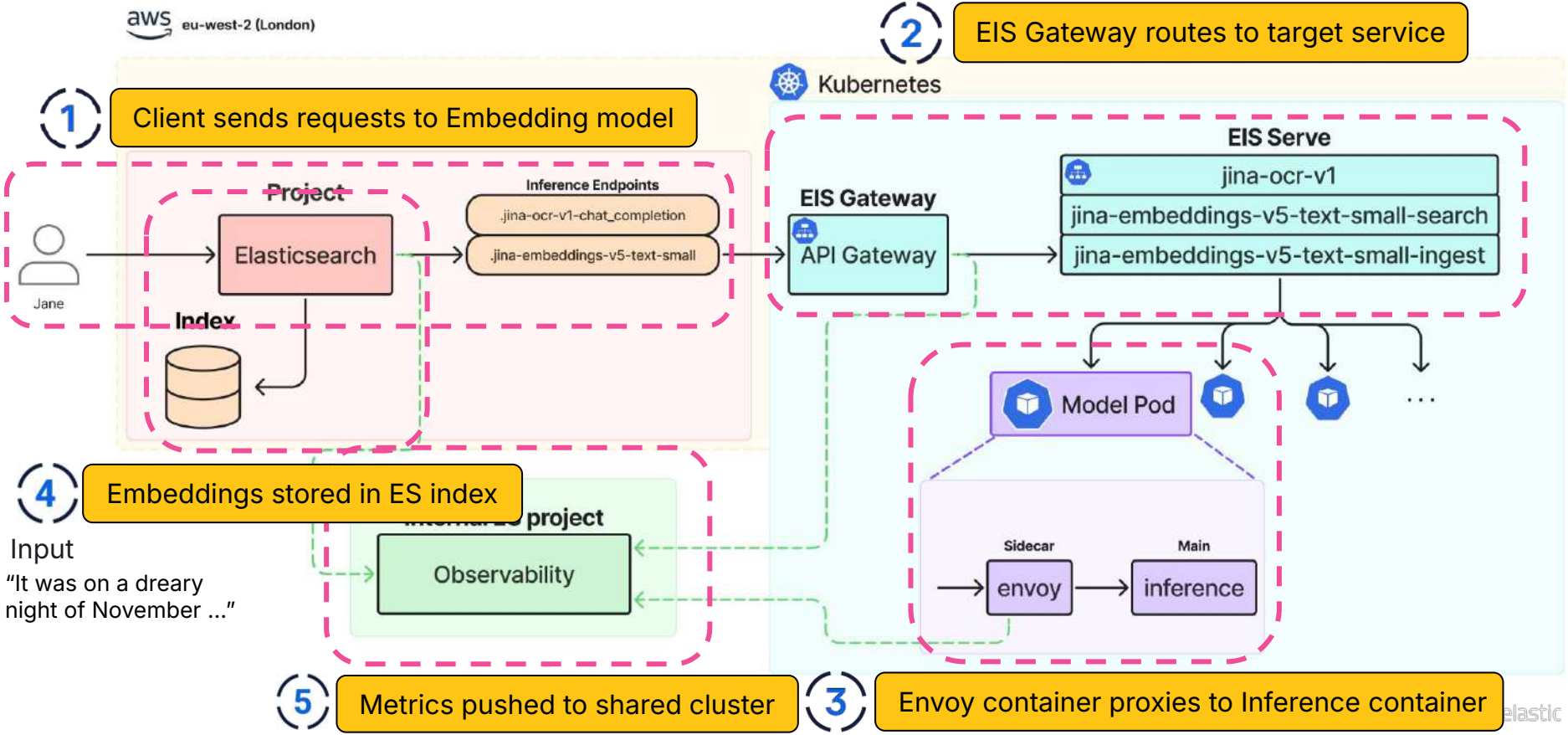
Managed Platforms

Pipeline

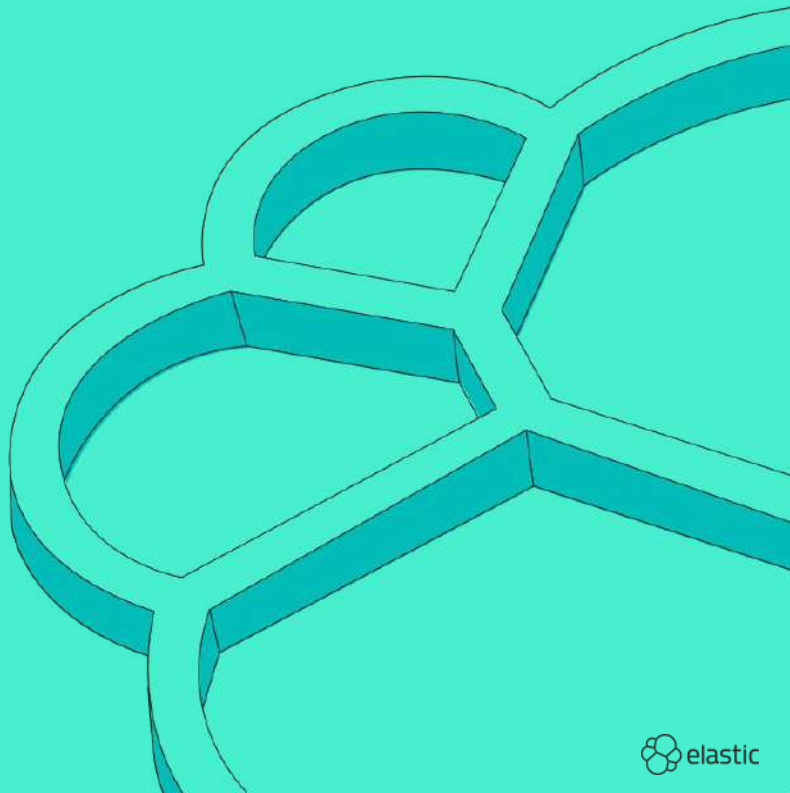


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Diagram - Ingest



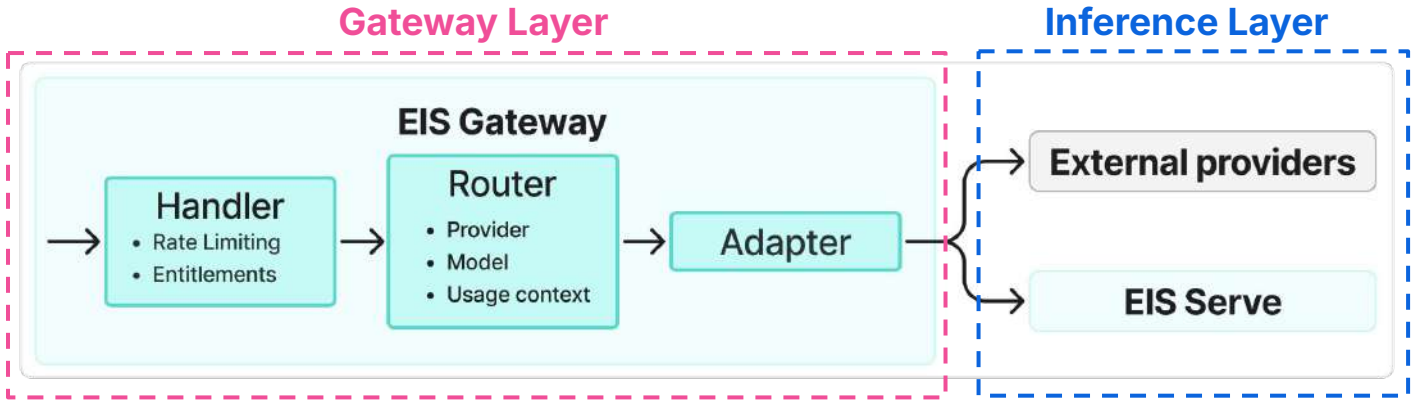
Demo: Ingest



Isolated layers

Key points

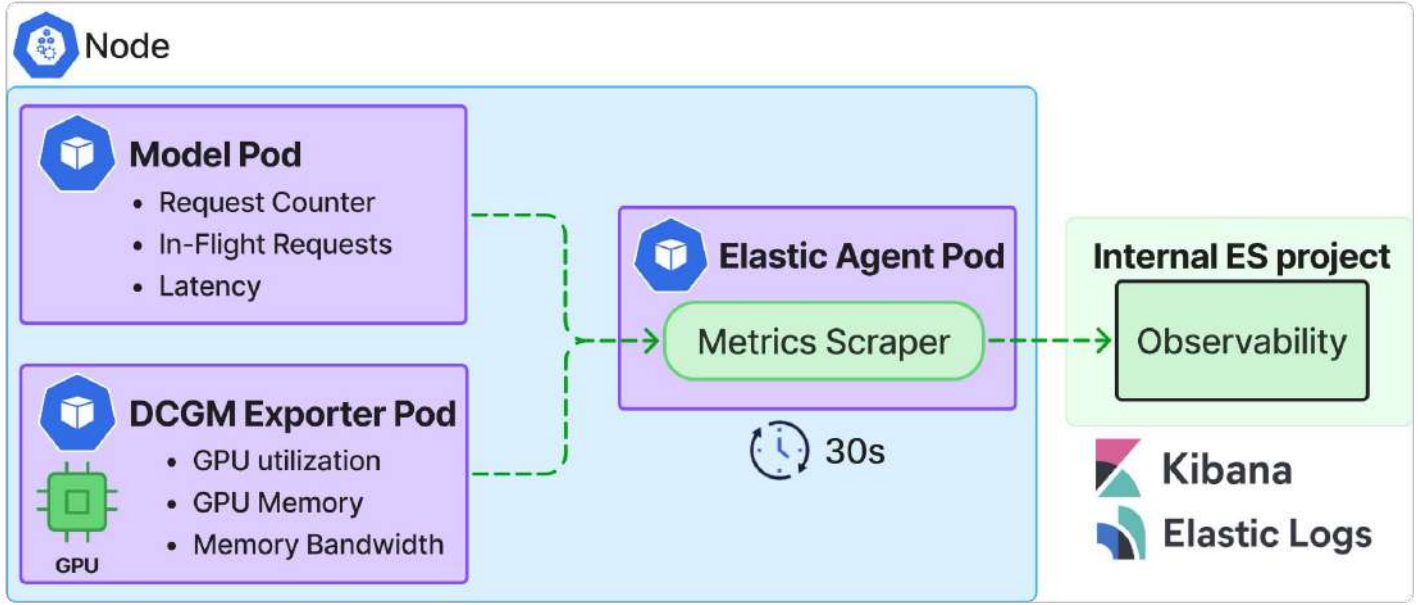
Inference server



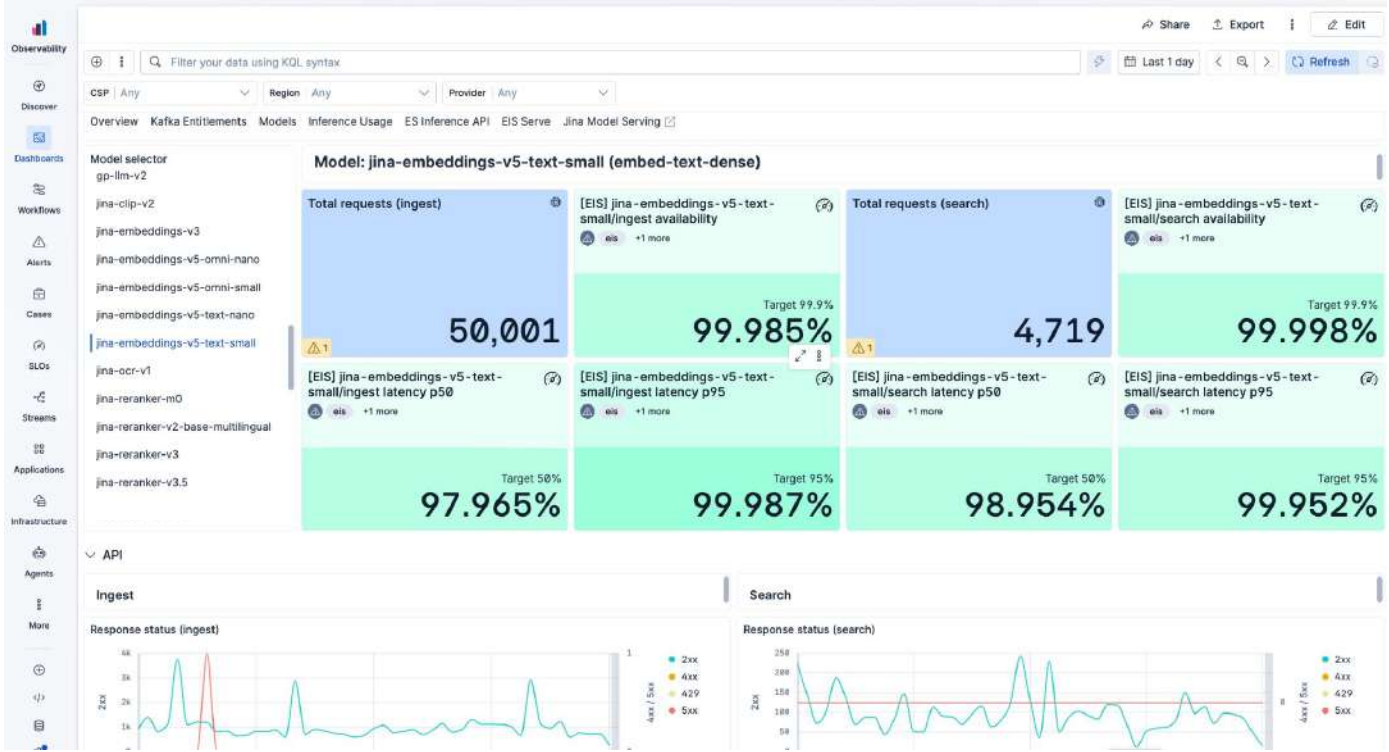
Observability

Key points

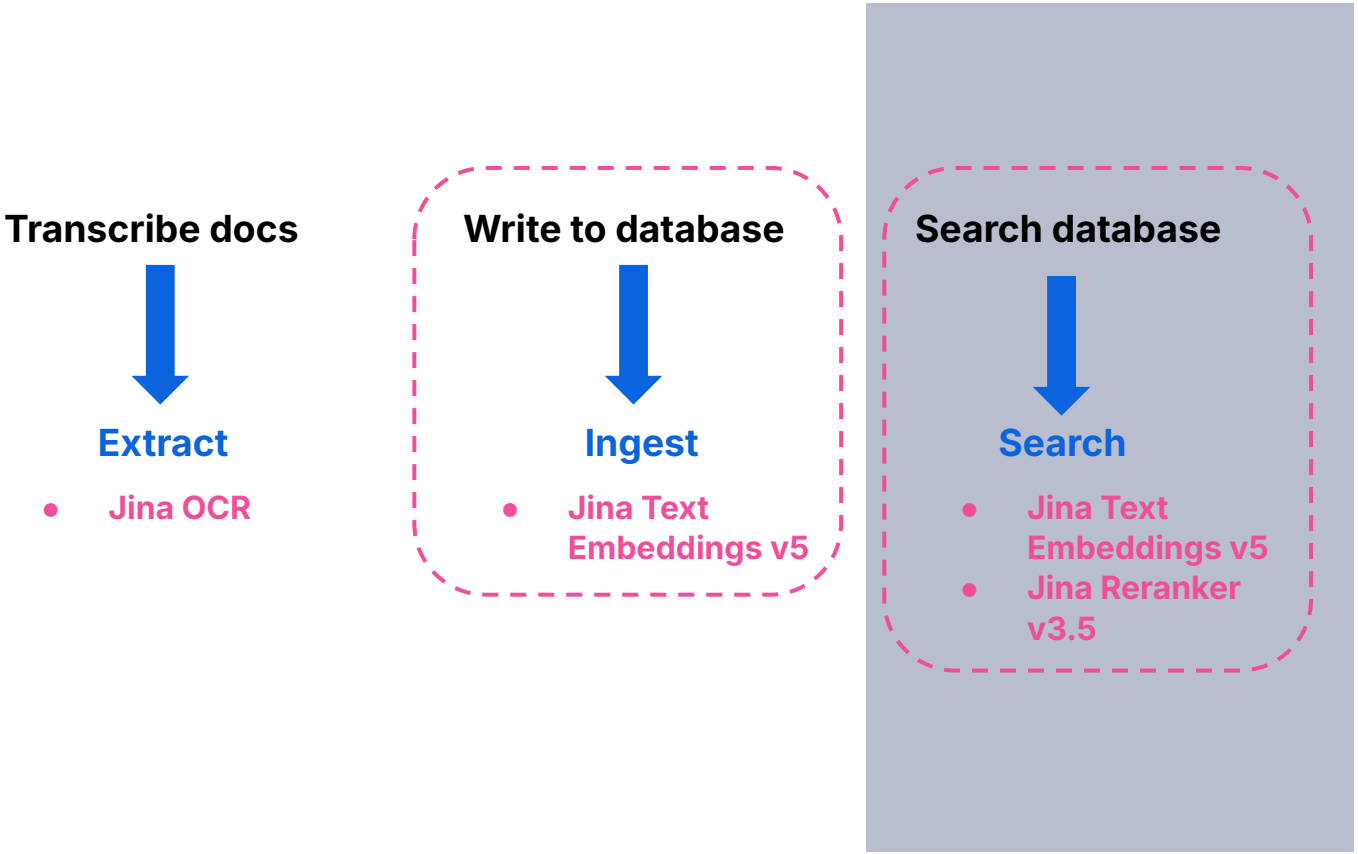
Inference server



Dashboard

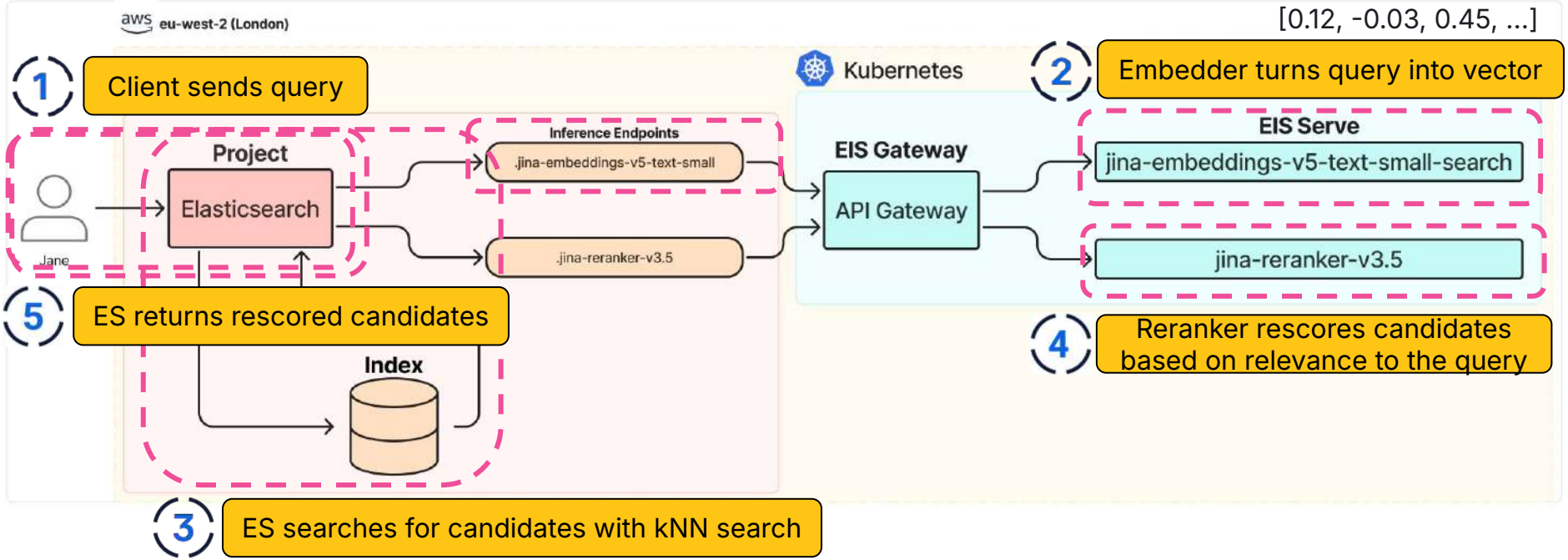

Observability
Discover
Dashboards
Workflows
Alerts
Cases
SLOs
Streams
Applications
Infrastructure
Agents
More
Share
Export
Edit
Filter your data using KQL syntax
Last 1 day
Refresh
CSP Any
Region Any
Provider Any
Overview
Kafka Entitlements
Models
Inference Usage
ES Inference API
EIS Serve
Jina Model Serving
Model selector
gp-llm-v2
jina-clip-v2
jina-embeddings-v3
jina-embeddings-v5-omni-nano
jina-embeddings-v5-omni-small
jina-embeddings-v5-text-nano
jina-embeddings-v5-text-small
jina-ocr-v1
jina-reranker-m0
jina-reranker-v2-base-multilingual
jina-reranker-v3
jina-reranker-v3.5
Model: jina-embeddings-v5-text-small (embed-text-dense)
Total requests (ingest)
50,001
[EIS] jina-embeddings-v5-text-small/ingest availability
Target 99.9%
99.985%
Total requests (search)
4,719
[EIS] jina-embeddings-v5-text-small/search availability
Target 99.9%
99.998%
[EIS] jina-embeddings-v5-text-small/ingest latency p50
Target 50%
97.965%
[EIS] jina-embeddings-v5-text-small/ingest latency p95
Target 95%
99.987%
[EIS] jina-embeddings-v5-text-small/search latency p50
Target 50%
98.954%
[EIS] jina-embeddings-v5-text-small/search latency p95
Target 95%
99.952%
API
Ingest
Search
Response status (ingest)
Response status (search)
2xx
4xx
429
5xx

Pipeline



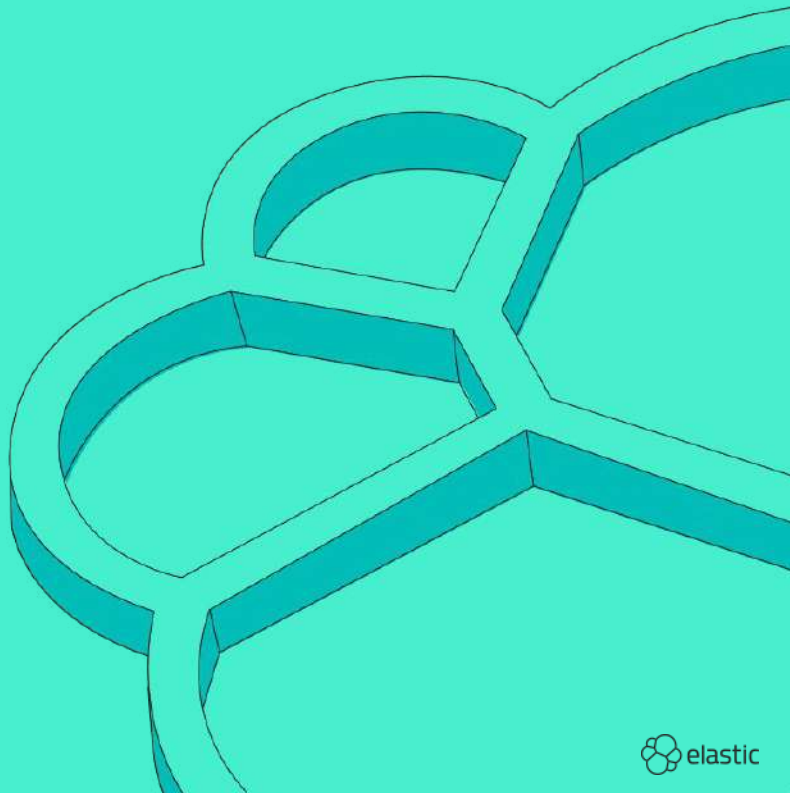
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Diagram - Search



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Demo: Search



Pipeline

Transcribe docs



Extract

- Jina OCR

Write to database



Ingest

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Search database



Search

- Jina Reranker v3.5
- Jina Text Embeddings v5



Users



Region



CSP



Time



Workload

How does EIS handle this in Production?

Key points

Performance at scale



Autoscaling

- Elastic Pod Autoscaler
  - In-flight requests
  - CPU utilization
  - vLLM queued requests



Scaling Config

- Estimate min. instance count
  - Region
  - Model
  - Deployment type
  - GPU
- Adjusted periodically



Workload Isolation

- Split deployment types
  - Search
  - Ingest



Optimizations

- vLLM
- Smart Batching

Conclusion

Smarter context

You don’t need to throw everything at an LLM. **Small models** provide better context for a fraction of the cost.

E2E on EIS

Elastic Inference Service has **all the tools you need** for visual intelligence and semantic retrieval.

Adapt to change

The AI landscape is changing fast. Make your platform as **flexible** as possible.

Scale first

Prioritise **scalability** when building your platform. **Robust observability** is key.

We're hiring!



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Thank you!



Q&A

